Didar ML Interview Task

Hello, dear candidate,

We want to congratulate you on passing the previous step of the interview, and now you are in the challenge step.

Objective

The task comes from a CRM company which tracks customer activity and runs customer-retention campaigns. In the past, account managers contacted a subset of customers.

Now, the CRM team has planned to contact ٥٠ customers during its next retention campaign. Using the provided data, develop a method for prioritizing those customers with the goal of maximizing customer retention and preventing their churn.

The dataset contains customer information, product usage before the contact decision, whether contact occurred, and whether the customer later churned.

Your solution should use both the customer-level features and the ٦٠-day product-usage history.

Data

١. **customer_data.csv**

Each row represents one customer and contains pre-contact features:

- **customer_id**: Customer identifier
- **company_size**: Number of employees in the customer's company
- **package**: Customer subscription type
- **tenure_months**: Number of months the customer has purchased the plan
- **Subscription_type**: Monthly or annual Subscription
- **monthly_revenue**: Monthly revenue generated by the customer
- **support_tickets_٣٠d**: Number of support tickets submitted in the previous ٣٠ days
- **email_engagement**: Customer email-engagement score between ٠ and ١
- **contacted**: Whether the customer received proactive contact
- **churned**: Whether the customer churned **after** the campaign decision date

٢. **usage_timeseries.csv**

   Contains ٦٠ days of product usage for each customer:
   - **customer_id**: Customer identifier
   - **day**: Day relative to the contact decision, from -٥٩ to ٠
   - **daily_usage**: Number of product-usage actions recorded that day

All features and usage values were recorded before the campaign decision date.

Deliverables

١. Reproducible Python code, preferably as a notebook, that:
   a. explores the supplied data
   b. shows insights about the data
   c. develops your prioritization approach
   d. produces a ranked list of ٥٠ customer IDs
٢. A **one-page** report explaining:
   a. your methodology
   b. how both data sources were used
   c. how you evaluated your approach
   d. the interpretation of your ranking score
   e. your level of confidence in the resulting ranking

Model complexity is not an evaluation criterion. We are primarily interested in the quality of your reasoning, methodology, evaluation, and interpretation.

Once you believe that you are done, please attach your results to an email and send it back in a reply.

Deadline: two days

Good Luck